

CreditWise Loan System

Intelligent ML-Powered Loan Approval · SecureTrust Bank

Automating credit decisions with speed, fairness and accuracy

1,000+
Records

20
Features

3
ML Models

Problem Statement

SecureTrust Bank offers personal and home loans to customers across urban and rural regions of India. Every day, hundreds of customers apply for loans through online and branch applications. Until now, loan officers manually evaluate each application by checking income proofs, employment details, credit history and other documents — a process that is **time-consuming, biased and inconsistent**.

Challenge 01

Good customers sometimes get **rejected**, causing loss of business.

Challenge 02

High-risk customers sometimes get **approved**, causing financial losses.

To solve this, SecureTrust Bank wants an **intelligent loan approval system** powered by Machine Learning that can automatically analyse applicant details and **predict whether a loan should be Approved or Rejected** before final human verification.

ML Pipeline Overview

1

Data Loading

Read 1,000 loan application records from CSV.

2

Missing Value Handling

Mean imputation for numerics; mode for categoricals.

3

EDA

Distribution plots, box-plots, correlation heatmap.

4

Encoding

Label encoding + One-Hot encoding for categoricals.

5

Feature Engineering

DTI^2 and $Credit\ Score^2$ polynomial features added.

6

Train / Test Split

80 / 20 stratified split with `random_state = 42`.

7	Scaling StandardScaler applied after split to prevent leakage.
8	Model Training & Evaluation Logistic Regression · KNN · Naive Bayes — Precision / Recall / F1.

Dataset Description

Each row represents a **loan applicant** and contains 19 input features describing their personal, financial and credit profile. The target variable is **Loan_Approved** (1 = Approved, 0 = Rejected).

Column	Type	Description
Applicant_ID	ID	Unique applicant identifier
Applicant_Income	Numeric	Monthly income of the applicant (INR)
Coapplicant_Income	Numeric	Monthly income of co-applicant (INR)
Employment_Status	Categorical	Salaried / Self-Employed / Business
Age	Numeric	Age of the applicant
Marital_Status	Categorical	Married / Single
Dependents	Numeric	Number of financial dependents
Credit_Score	Numeric	Credit bureau score
Existing_Loans	Numeric	Number of currently active loans
DTI_Ratio	Numeric	Debt-to-Income ratio
Savings	Numeric	Current savings balance (INR)
Collateral_Value	Numeric	Value of collateral provided (INR)
Loan_Amount	Numeric	Requested loan amount (INR)
Loan_Term	Numeric	Loan duration in months
Loan_Purpose	Categorical	Home / Education / Personal / Business
Property_Area	Categorical	Urban / Semi-Urban / Rural
Education_Level	Categorical	Graduate / Postgraduate / Undergraduate
Gender	Categorical	Male / Female
Employer_Category	Categorical	Govt / Private / Self
Loan_Approved	Target	1 = Approved, 0 = Rejected

Models & Technology Stack

Model	Key Strengths	Use Case Fit
Logistic Regression	Interpretable, fast, strong baseline	Understand feature weights
K-Nearest Neighbours	Non-parametric, intuitive	Similarity-based decisions
Naive Bayes	Best Precision — selected model	Fast probabilistic inference

Tech Stack

Python 3	Pandas	NumPy	Scikit-learn	Seaborn	Matplotlib	Jupyter
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Key Insights from EDA

Credit Score	Strongest single predictor of loan approval across all models.
DTI Ratio	Higher debt-to-income ratio significantly reduces approval probability.
Savings	Applicants with higher savings balances show better approval rates.
Feature Eng.	Squared DTI and Credit Score features improved model precision.
Class Balance	Dataset is moderately balanced; no aggressive resampling needed.
Best Model	Naive Bayes achieved highest Precision after feature engineering.